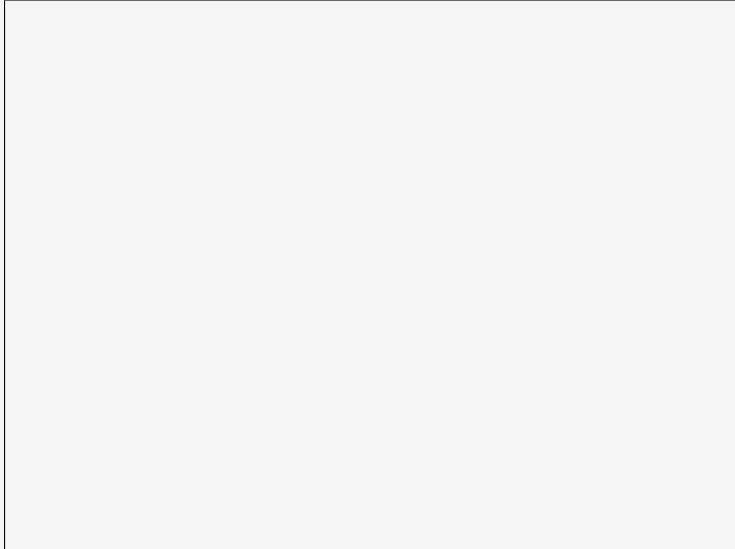


Wait, who was that? Tracking interacting trajectories with label uncertainty

Henry Scharf
Department of Mathematics
University of Arizona

June 30, 2026



Social Behavior



Pitman and Durban (2012). Cooperative hunting behavior... *Marine Mammal Science*.



Photo Credit: John Durban

Social Behavior

Research questions

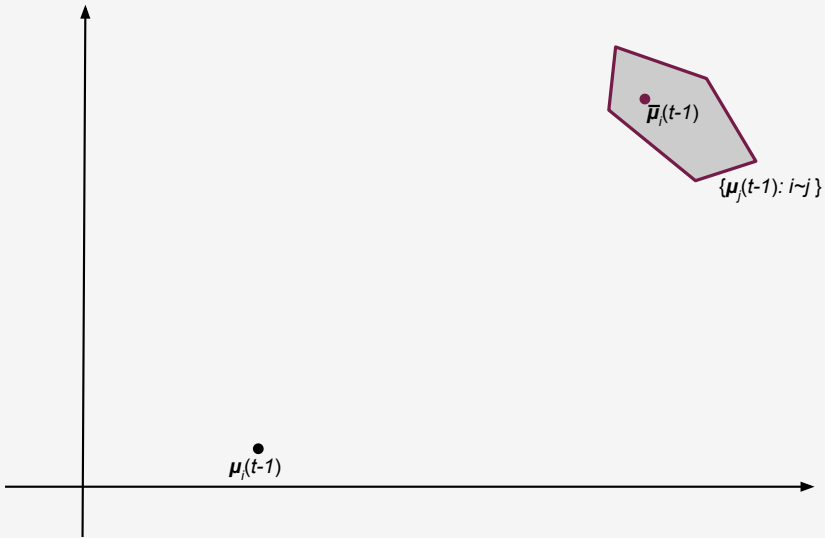
- How do social connections affect behavior?
- How might those connections change over time?
- How do environmental conditions impact social connections?

Challenges

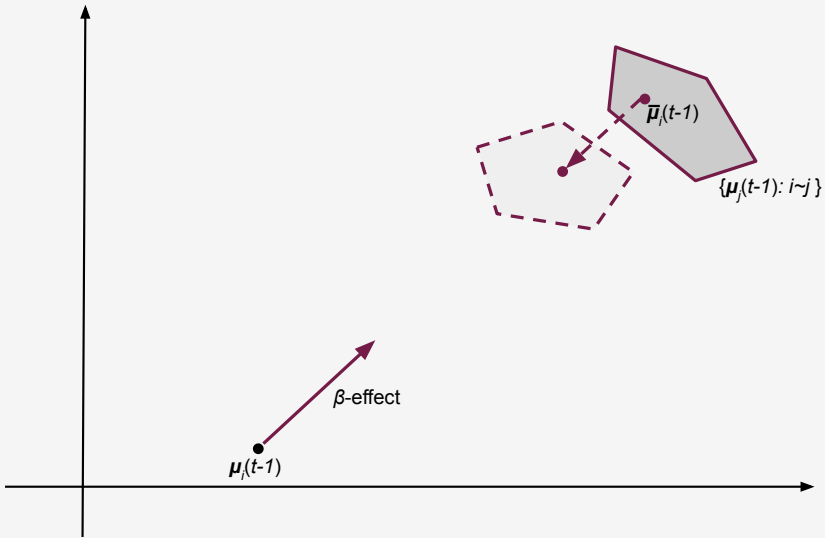
- Observing animals can be intrusive, opportunistic, and expensive.
- Animals often occupy habitat inaccessible to humans.
- Definitions of meaningful interactions are *ad hoc* based on individual studies.

A Social Model for Movement

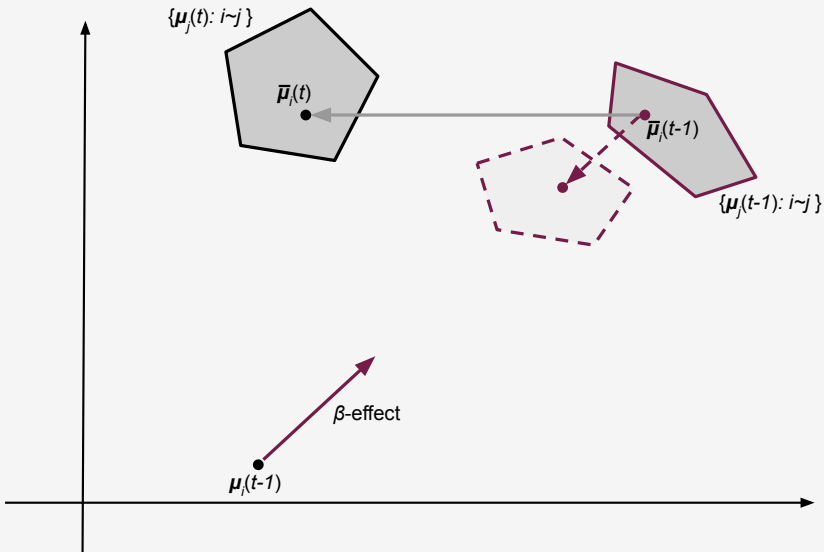
Attraction & Alignment



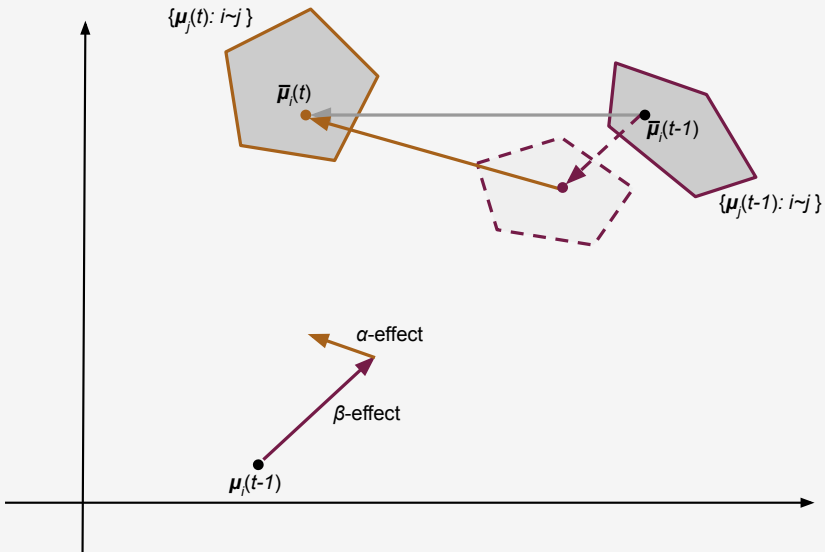
Attraction & Alignment



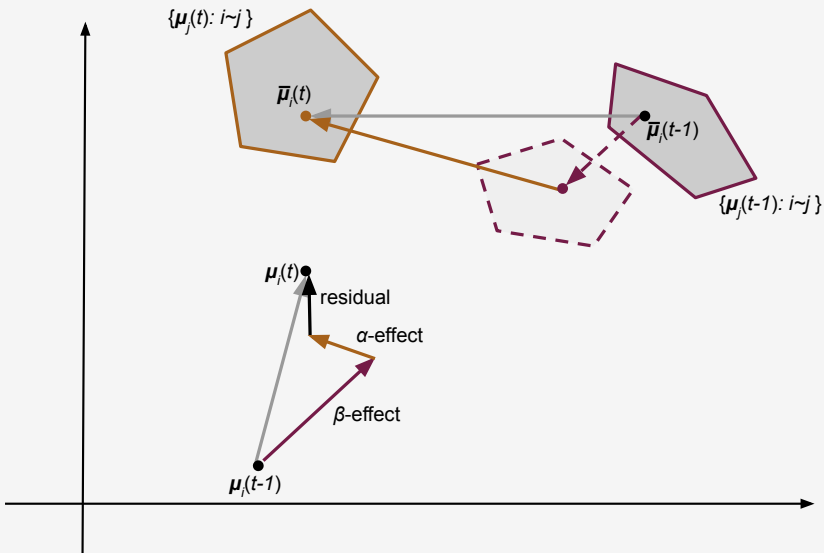
Attraction & Alignment



Attraction & Alignment



Attraction & Alignment



State-Space Model for Social Movement

$$\mathbf{y}(t) | \boldsymbol{\mu}(t) = \boldsymbol{\mu}(t) + \boldsymbol{\varepsilon}(t),$$

$$\boldsymbol{\varepsilon}(t) \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$$

$$\boldsymbol{\mu}(t) | \boldsymbol{\mu}(t-1) = \mathbf{A}(t) \boldsymbol{\mu}(t-1) + \boldsymbol{\eta}(t),$$

$$\boldsymbol{\eta}(t) \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{Q}^{-1}(t))$$

$$\mathbf{A}_{ij}(t) = \begin{cases} \beta(t) \frac{w_{ij}(t)}{w_{i+}(t)}, & i \neq j \\ 1 - \beta(t), & i = j \end{cases}$$

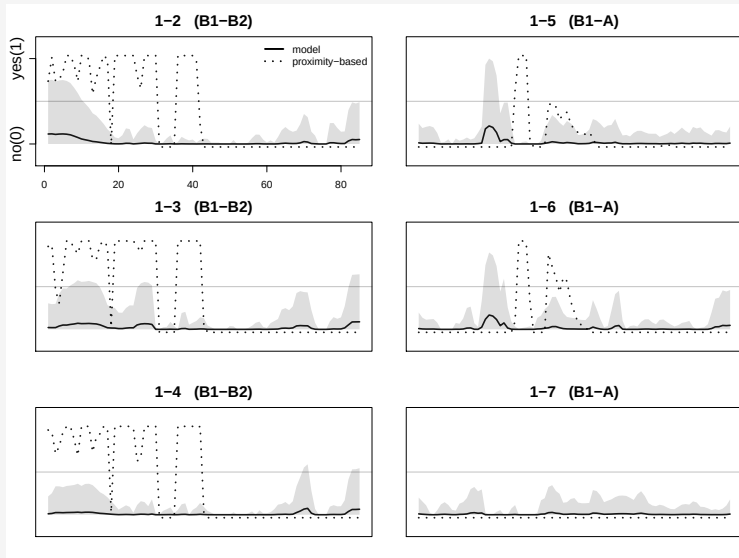
$$\mathbf{Q}_{ij}(t) = \begin{cases} -\alpha(t) w_{ij}(t), & i \neq j \\ w_{i+}(t), & i = j \end{cases}$$

$$\mathbf{A}(t) = \mathbf{I} + \beta(t)(\overline{\mathbf{W}}(t) - \mathbf{I})$$

- Key reason for SSMs: Kalman (particle) filters marginalize over state process.

Application: Killer Whales

- $\alpha(t), \beta(t)$ constant in time.
- Stationary Markov process prior for $w_{ij}(t)$.
- Connections defined purely by distance are spurious.



Drone-based movement data: Opportunities and obstacles

Drone-based Movement Data

- + High spatial/temporal resolution
- + Concurrent observations for social behavior
- + Often minimally intrusive
- Complex observation process
- Short duration
- Requires local pilot



(a) Wouldn't it be great if they were all like this?

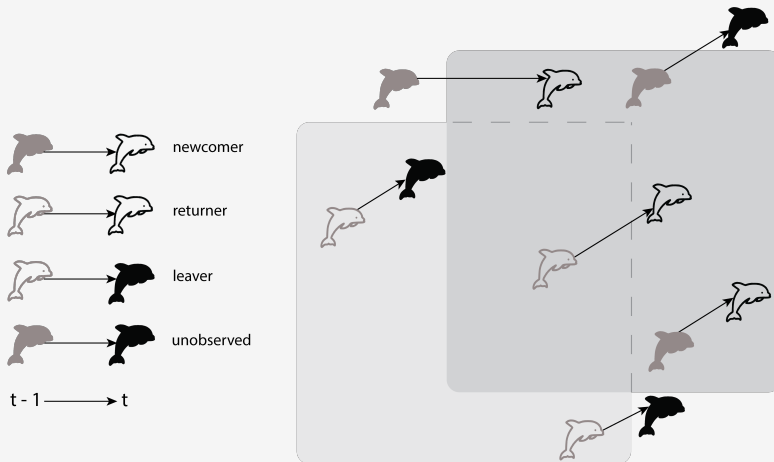


(b) Yeah, well...

Photo Credit: Samantha Leander

Label Uncertainty

- Newcomers must be assigned a unique id (e.g., a 30-minute observation window resulted in 104 unique labels applied to a population of 18 dolphins).



Confine Attention to the Returners

- Let $\mathbf{x}(t)$ be a vector of covariates measured at time t .

$$\text{logit}(\alpha(t)) = \mathbf{x}'(t)\boldsymbol{\delta}_\alpha, \quad \text{logit}(\beta(t)) = \mathbf{x}'(t)\boldsymbol{\delta}_\beta$$

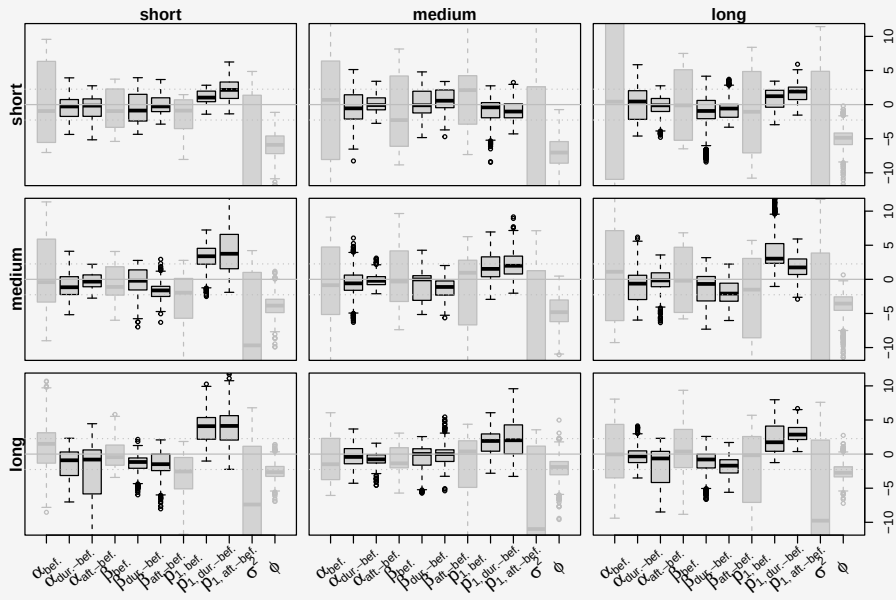
- Heuristically: assume that, at any given moment, the entire population of animals is limited to the returners.

$$L(\boldsymbol{\theta} | \mathbf{W}, \boldsymbol{\mu}) = \prod_{t=2}^T \text{N}(\boldsymbol{\mu}(t); \mathbf{A}(t-1)\boldsymbol{\mu}(t-1), \sigma^2 \mathbf{Q}^{-1}(t-1)),$$

$$\tilde{L}(\boldsymbol{\theta} | \mathbf{W}_{\text{ret}}, \boldsymbol{\mu}_{\text{ret}}) = \prod_{t=2}^T \text{N}(\boldsymbol{\mu}_{\text{ret}}(t); \mathbf{A}_{\text{ret}}(t-1)\boldsymbol{\mu}_{\text{ret}}(t-1), \sigma^2 \mathbf{Q}_{\text{ret}}^{-1}(t-1))$$

- Brute force, naive, myopic, data wasteful...but computable.

A Simulation Study of Bias



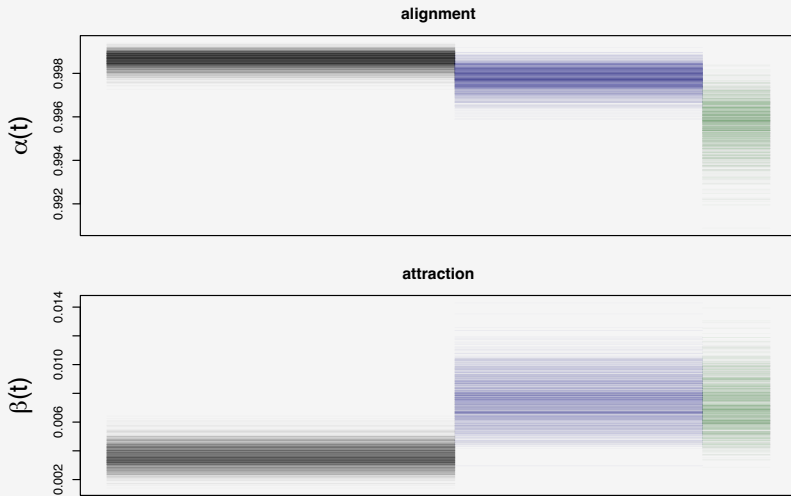
Application: Risso's dolphins

- A controlled exposure experiment was observed with approximately 14 minutes of movement before, 10 minutes during, and 2.5 minutes after sonar.
- 18 individuals were observed with 104 unique labels. At most 9 individuals were observed simultaneously.
- Drones recorded images once per second.
- Quartiles for number of consecutive observations: 7, 26, 74.
- Samples from posterior using MCMC implemented with NIMBLE.

Boulil et al. (2023). Detecting changes in dynamic social networks using multiply-labeled movement data. *JABES*.

de Valpine et al. (2017). Programming with models: Writing statistical algorithms for general model structures with NIMBLE. *Journal of Computational and Graphical Statistics*

Application: Risso's dolphins



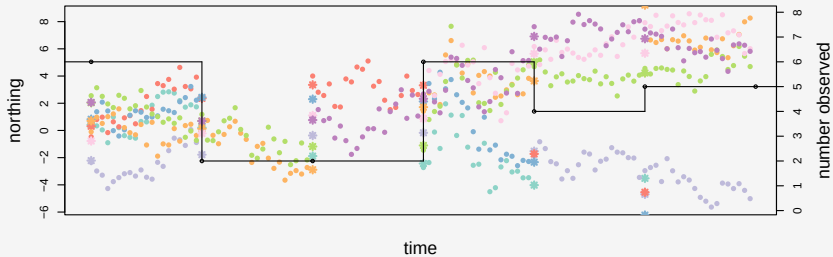
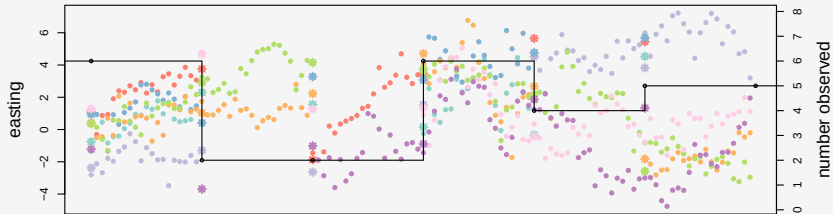
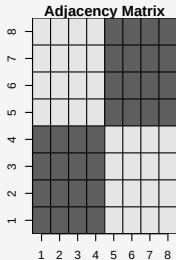
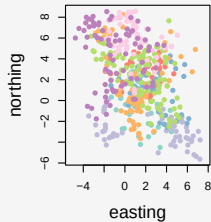
State-Space Model for Social Movement: Uncertain Labels

$$\begin{aligned} \mathbf{y}(t) | \boldsymbol{\mu}(t) &= \mathbf{Z}(t) \boldsymbol{\mu}(t) + \boldsymbol{\varepsilon}(t), & \boldsymbol{\varepsilon}(t) &\sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}) \\ \boldsymbol{\mu}(t) | \boldsymbol{\mu}(t-1) &= \mathbf{A}(t) \boldsymbol{\mu}(t-1) + \boldsymbol{\eta}(t), & \boldsymbol{\eta}(t) &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{Q}^{-1}(t)) \end{aligned}$$

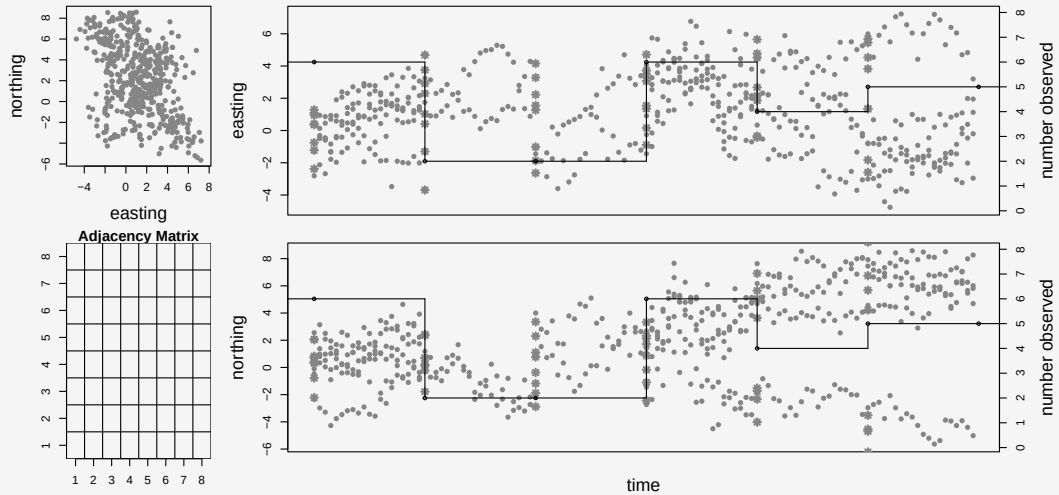
$$\begin{aligned} z_{ij}(t) &= \mathbf{1}_{\{\mathbb{E}[y_i(t)] = \mu_j(t)\}} \\ \mathbf{z}_i(t) | \mathbf{z}_{<i}(t), \mathbf{z}(t-1) &\sim \text{Multi}(\pi_i(t)) \\ \pi_1(t) &= (1, \dots, 1) = \mathbf{1}_J \\ \pi_i(t) &= \pi_{i-1}(t) \odot (\mathbf{1}_J - \mathbf{z}_{i-1}(t)), & i &= 2, \dots, J \end{aligned}$$

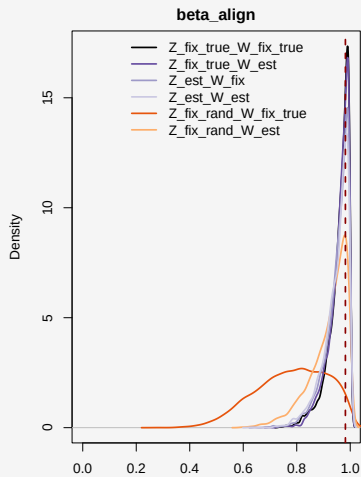
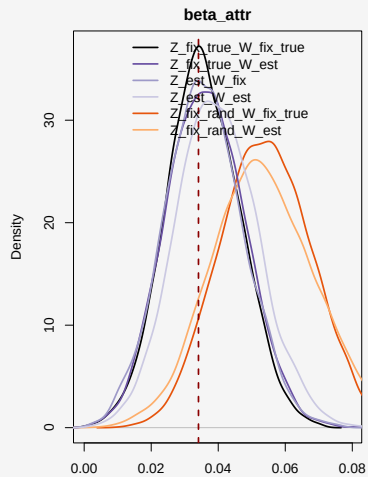
- Key reason for SSMs: Kalman (particle) filters marginalize over state process.
- *Partial* marginalization: Update the state, $\boldsymbol{\mu}(t)$, only at times when $\mathbf{Z}(t)$ changes.

State-Space Model for Social Movement: Uncertain Labels



State-Space Model for Social Movement: Uncertain Labels





Thanks!

Killer Whales

John Durban, SEA Inc.

Josh London, NOAA

Mevin Hooten, UT Austin

Bailey Fosdick, GTI Energy, CO School of Public Health

Risso's Dolphins

John Durban, Brandon Southall, Samantha G. M.

Leander, SEA Inc.

Holly Fearnbach, SR³ SeaLife

Response, Rehabilitation and Research

Trevor Joyce, Environmental Assessment Services

Students and Postdocs

Anisha Mathias, University of Arizona

ZaineB Boulil, SDSU



SDSU

San Diego State University



THE UNIVERSITY OF ARIZONA
COLLEGE OF SCIENCE

Mathematics

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- Scharf, H. R. and Buderman, F. E. (2020). Animal movement models for multiple individuals. *Wiley Interdisciplinary Reviews: Computational Statistics*, 12(6):e1506.
- Scharf, H. R., Hooten, M. B., Fosdick, B. K., Johnson, D. S., London, J. M., and Durban, J. W. (2016). Dynamic social networks based on movement. *The Annals of Applied Statistics*, 10(4):2182–2202.

Movement process

$$\bar{\mu}_i(t) = \frac{1}{w_{i+}(t)} \sum_{j \neq i} w_{ij}(t) \mu_j(t) \quad \text{centroid}$$

$$w_{i+}(t) = \sum_j w_{ij}(t)$$

$$\tilde{\mu}_i(t) = \bar{\mu}_i(t) - \mu_i(t) \quad \text{direction of attraction}$$

Movement process

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$$\tilde{\mu}_i^*(t) = \frac{\bar{\mu}_i(t) - \mu_i(t)}{\|\bar{\mu}_i(t) - \mu_i(t)\|_2} \quad \text{absolute interpretation of } \beta$$

Movement process

$$\begin{aligned}\mathbb{E} [\boldsymbol{\mu}_i(t) | \boldsymbol{\mu}_{-i}(t), \boldsymbol{\mu}(t-1)] &= \underbrace{\boldsymbol{\mu}_i(t-1)}_{\text{previous location}} + \underbrace{\beta \tilde{\boldsymbol{\mu}}_i(t-1)}_{\text{attraction}} \\ &+ \underbrace{\alpha \sum_{j \neq i} \frac{w_{ij}(t-1)}{w_{i+}(t-1)} [\boldsymbol{\mu}_j(t) - (\boldsymbol{\mu}_j(t-1) + \beta \tilde{\boldsymbol{\mu}}_j(t-1))]}_{\text{alignment}}\end{aligned}$$

Movement process

$$\begin{aligned}\mathbb{E} [\boldsymbol{\mu}_i(t) | \boldsymbol{\mu}_{-i}(t), \boldsymbol{\mu}(t-1)] &= \underbrace{\boldsymbol{\mu}_i(t-1)}_{\text{previous location}} + \underbrace{\beta \tilde{\boldsymbol{\mu}}_i(t-1)}_{\text{attraction}} \\ &+ \underbrace{\alpha \sum_{j \neq i} \frac{w_{ij}(t-1)}{w_{i+}(t-1)} [\boldsymbol{\mu}_j(t) - (\boldsymbol{\mu}_j(t-1) + \beta \tilde{\boldsymbol{\mu}}_j(t-1))]}_{\text{alignment}}\end{aligned}$$

$$Q_{ij}(t) = \begin{cases} -\alpha w_{ij}(t), & i \neq j \\ w_{i+}(t), & i = j \end{cases}$$

$$\boldsymbol{\mu}(t) | \boldsymbol{\mu}(t-1) \sim \text{N}(\boldsymbol{\mu}(t-1) + \beta \tilde{\boldsymbol{\mu}}(t-1), \sigma^2 \mathbf{Q}^{-1}(t-1))$$